
Mobile Product Segmentation & Recommendation System

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Project 4

Mobile Product Segmentation & Recommendation System

Project Documentation

1. Project Overview

A Streamlit-based Mobile Product Segmentation & Recommendation System that analyzes processed mobile product data, displays dashboard information, performs cluster analysis, and recommends similar mobile products based on price, ratings, and specifications.

2. Technologies Used

- Python
- Streamlit
- Pandas
- Joblib
- Scikit-learn
- PCA
- K-Means
- Nearest Neighbors

3. Main Features

- Dashboard with total records, total columns, and brand distribution.
- Cluster distribution and cluster characteristics.
- Cluster insights for four identified clusters.
- PCA-based two-dimensional cluster visualization.
- Mobile product selection for recommendations.
- Similar product recommendations using a saved nearest-neighbor model.

4. Navigation Pages

- Dashboard – Dataset overview and brand distribution.
- Cluster Analysis – Cluster distribution, characteristics, insights, and PCA visualization.
- Recommendations – Select a mobile product and view similar recommended products.

5. Recommendation Features

- price_usd

- rating
- battery_life_rating
- camera_rating
- performance_rating
- design_rating
- display_rating

6. Cluster Features

- price_usd
- rating
- battery_life_rating
- camera_rating
- performance_rating
- design_rating
- display_rating
- helpful_votes

7. Cluster Insights

- Cluster 0 – Mid-range / Average Products
- Cluster 1 – Premium Products
- Cluster 2 – Low-rated / Poor-performing Products
- Cluster 3 – High-performing / Highly Rated Products

8. Required Files

- app.py
- mobile_processed.csv
- kmeans.pkl
- nn_model.pkl
- scaler.pkl
- recommendation_scaler.pkl

9. How to Run

1. Install Python.
2. Place app.py, mobile_processed.csv, and all required .pkl files in the same project folder.
3. Install the required Python packages.
4. Open a terminal in the project folder.
5. Run: streamlit run app.py
6. Open the Streamlit URL displayed in the terminal.

10. Project Structure

Mobile-Product-Segmentation-Recommendation-System/

- └─ app.py
- └─ mobile_processed.csv
- └─ kmeans.pkl
- └─ nn_model.pkl
- └─ scaler.pkl
- └─ recommendation_scaler.pkl
- └─ README.md

11. Screenshots

Added screenshots of the Dashboard, Cluster Analysis, PCA visualization, and Recommendations pages here.

```
[59]: mobile_df[['sentiment','source']].isnull().sum()
```

```
[59]: sentiment    0  
      source      0  
      dtype: int64
```

```
[60]: # Check duplicates
```

```
[61]: mobile_df.duplicated().sum()
```

```
[61]: np.int64(0)
```

```
85]: # Step 1: Count brands
```

```
86]: brand_counts = mobile_df['brand'].value_counts()  
      brand_counts
```

```
86]: brand  
      Xiaomi      7241  
      Google      7234  
      Apple       7144  
      OnePlus     7136  
      Realme      7132  
      Motorola    7061  
      Samsung     7052  
      Name: count, dtype: int64
```

```
87]: # Step 2: Visualize Top Brands
```

```
89]: plt.figure(figsize=(5,5))  
  
      sns.barplot(  
          x=brand_counts.index,  
          y=brand_counts.values  
      )  
  
      plt.xticks(rotation=90)  
  
      plt.xlabel("Brand")  
  
      plt.ylabel("Number of Reviews")  
  
      plt.title("Mobile Brand Distribution")  
  
      plt.show()
```

```

3]: # Test one product

3]: distances, indices = nn_model.kneighbors(
    X_recommend_scaled[0].reshape(1, -1)
)

3]: indices

3]: array([[ 0, 16761, 37247, 28925, 45398, 5956]])

1]: distances

1]: array([[0.          , 0.00015042, 0.00321    , 0.00332127, 0.00408469,
          0.00429529]])

]:

128]: kmeans.fit(X_scaled)

128]: ▾ KMeans ⓘ ⓘ
    ► Parameters

129]: # Step 4 – Get cluster labels

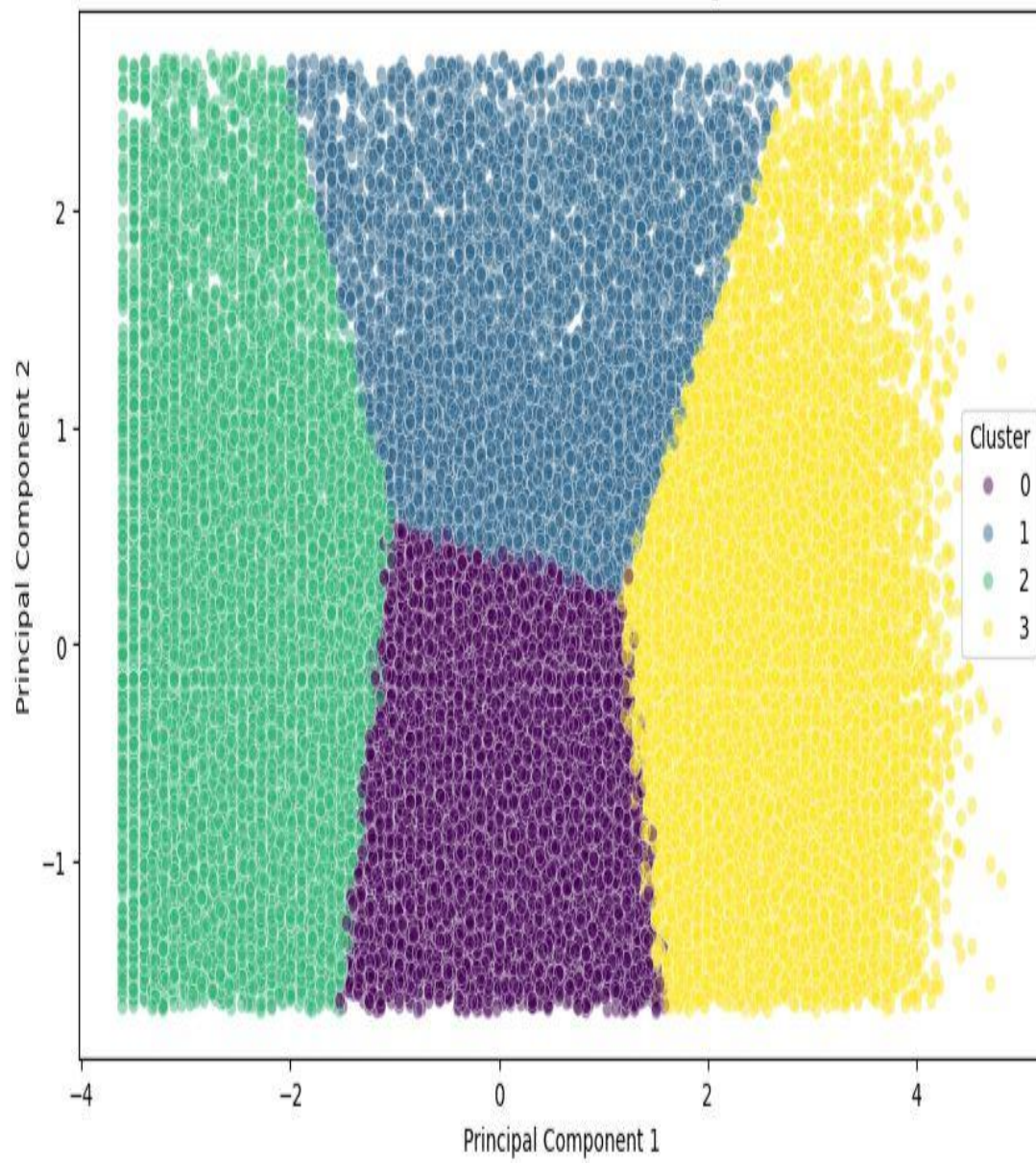
130]: cluster_labels = kmeans.labels_

131]: cluster_labels[:20]

131]: array([2, 0, 3, 2, 2, 3, 0, 1, 0, 0, 0, 2, 3, 2, 1, 2, 3, 2, 3,
          dtype=int32)

```

K-Means Cluster Visualization using PCA





Mobile Product Segmentation & Recommendation System



Product Recommendation

Select a mobile product to find similar products based on price, ratings, and specifications.

Select a mobile product

Apple - iPhone SE



Get Recommendations

Selected Product

Apple iPhone SE

Price: \$1155.32 | Rating: 3.0

Recommended Products

	brand	model	price_usd	rating	battery_life_rating	camera_rating	performance_rating
36508	Apple	iPhone 13	1041.07	3	2	3	3
22091	Google	Pixel 8	1060.84	3	2	4	4
12137	Apple	iPhone 14	1352.62	4	2	3	4
21672	Samsung	Galaxy A55	1026.81	3	2	3	3
341	Samsung	Galaxy Z Flip	1267.06	3	3	3	4